

RESEARCH ARTICLE

Geographical distribution and diversity of maize (*Zea mays* L. subsp. *mays*) races in Mexico

Quetzalcóatl Orozco-Ramírez · Hugo Perales · Robert J. Hijmans

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Abstract The center of origin of maize is in Mexico and maize diversity is very high in many parts of the country. Here we describe and analyze the geographic distribution and diversity of maize using a database of 18,176 georeferenced observations of maize races. The number of observations by race follows the common “hollow curve” distribution, with six races comprising 54 % of the records and 37 races only 10 %. Few races are widespread, and some have a highly localized distribution. The average maximum distance ($D_{\max}$) between two records of the same race was 1169 km, but seven races had a $D_{\max}$ lower than 200 km. The average circular area range size (CA_{25})

was 51, but 14 races had a CA_{25} lower than 10. Some races have very few observations but were found over a large area. These races may not be valid and their classification should be reviewed. Areas of high race richness and diversity are in mountainous northwest, south central, and southern Mexico in the states of Oaxaca, Jalisco, Michoacán and Chiapas. Six contiguous maize regions were defined using hierarchical clustering. Despite the high sample size, further sampling could improve estimates of race diversity in several regions.

Keywords Diversity mapping · Evenness · Maize races · Mexico · Richness · *Zea mays*

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Q. Orozco-Ramírez (✉)
Centro de Investigaciones en Geografía Ambiental,
UNAM, Antigua Carretera a Pátzcuaro No. 8701,
Col. Ex-Hacienda de San José de la Huerta,
58190 Morelia, Michoacán, Mexico
e-mail: qorozco@gmail.com

H. Perales
El Colegio de la Frontera Sur, Carretera Panamericana y
Periférico Sur s/n Barrio María Auxiliadora,
San Cristóbal de las Casas, Chiapas, Mexico

R. J. Hijmans
Department of Environmental Science and Policy,
University of California, One Shields Avenue, Davis,
CA 95616, USA

Introduction

Geographic patterns in the diversity of crop species have been studied to identify the areas of origin of crops, their dispersal from these areas, and to support collection and conservation (e.g., De Candolle 1892; Vavilov 1951; Harlan 1975; Brush 2004; Van Etten and Hijmans 2010; Kraft et al. 2014). Understanding these patterns can also help elucidate the past, current and future role of crop biodiversity in sustaining agriculture (Hawkes 1983; Bellon 1996; Brush 2000; Jarvis et al. 2008). Nevertheless, providing a comprehensive description of the geographic patterns of diversity in a particular crop remains a challenge due

to small and biased samples that are generally available. In contrast to studies of crop wild relatives (e.g. Hijmans and Spooner 2001; Jarvis et al. 2003), studies of infra specific crop diversity have mostly focused on relative small areas (Zimmerer and Douches 1991; Brush et al. 1995; van Etten et al. 2008) because of the lack of comprehensive and consistent infra-specific diversity data. Here we describe the distribution and diversity of maize in Mexico based on a large national level database.

Maize (*Zea mays* L. subsp. *mays*) originated in Mexico (Matsuoka et al. 2002), and Mexican farmers continue to grow a highly diverse set of maize genotypes (Hernández and Alanís 1970; Ortega-Paczka 1973; Vigouroux et al. 2008; Kato et al. 2009). Crop diversity can be expressed in several dimensions, including taxonomic, genetic, morphological, agronomic and usage traits variation. Mexican maize types are commonly grouped in ‘races’ (Anderson and Cutler 1942). For example, Wellhausen et al. (1952) used agronomical and morphological characteristics to distinguish 25 Mexican maize races; Sanchez et al. (2000) also used genetic criteria to define races. Although race is not a formal taxonomic category it is considered a useful unit to study the distribution of maize diversity in Mexico (Matsuoka et al. 2002; Vigouroux et al. 2008). Some races are associated with particular uses (Ortega-Paczka 2003) and with cultural groups (Perales et al. 2005; Brush and Perales 2007). Some races are rare, for example, Jala is only grown in a small valley in the state of Nayarit, whereas others, such as Tuxpeño, are widespread (Wellhausen et al. 1952). Maize is an out-breeding species and in farmers’ fields one can find plants that share traits of two or more races. Nevertheless, most Mexican maize races are well established in the sense that there is little dispute about their classification, morphology and traits. According to the most recent comprehensive treatment (Sanchez et al. 2000), there are 59 Mexican maize races, but a national maize survey (CONABIO 2013) reported 64 races.

In this paper we describe patterns of diversity in maize in Mexico using a unique national survey of 18,355 observations (CONABIO 2013). These data have previously been used to associate races with climate variables (Ruiz Corral et al. 2008) and to predict the entire potential distribution of particular races (Ureta et al. 2013). Ruiz Corral et al. (2013) used these data to build species distribution models to

predict ranges and ecological variables associated to these and Perales and Golicher (2014) mapped geographic patterns of potential maize race richness and proposed biogeographic regions for Mexico based on racial composition. This paper also maps patterns of race richness, but we focus on analyzing the observed data at different spatial resolutions and we provide additional measures of diversity. Information about maize diversity distribution is relevant for genetic resources conservation, maize dispersion, diversification and evolution.

Methods

Maize race data

Maize race distribution data for Mexico was taken from the Global Maize Project database (CONABIO 2013). This database is a compilation of historical records, including genebanks, and more recent surveys by several Mexican institutions coordinated by CONABIO (Comisión Nacional para el Conocimiento y Uso de la Biodiversidad; National Commission for the Understanding and Use of Biodiversity). The most common survey method has been to collect the largest number of maize distinct types within a village or other small region (Hernández 1972); but, particularly in the older records, collections have been made using different protocols. The database contains 22,931 records of maize observations, but for 4119 the race was not determined and an additional 455 records did not have geographic coordinates for the locations. There were 64 distinct race names in the database but we only used the 59 races listed by Sanchez et al. (2000). The races excluded from the data set were: Chiquito (43 observations), Complejo Serrano de Jalisco (18), Cubano Amarillo (65), Mushito de Michoacán (54), and Quicheño (1). The final version of the database we used contained 18,176 records (see supplemental material 2 Appendix S.3). All records included geographical coordinates verified by CONABIO, most also had elevation data, year of collection, primary race, and collector. Most observations (60 %) in this subset were recorded from 1997 to 2010, but there are records since 1927 and there were no records for 1985–1996. We evaluated leaving out the earliest records, as these races may no longer be present in some of these locations, but we decided to keep them

all in as that appeared to give the most robust patterns. Nevertheless, future work could attempt to use these data to investigate changes in maize diversity over time (cf. Dyer et al. 2014).

Range size

We computed two range size statistics: maximum distance ($D_{\max}$) and circular area (CA_r) (Hijmans and Spooner 2001) for each race to analyze maize race distribution. $D_{\max}$ is the largest distance (km) between any pair of observations of the same race. The circular area statistic is the area covered by circles of radius r around each point (areas that overlap are included once) divided by the area of a single circle. Because the area is relative to a circle, it is easily comparable across different studies and scales (radii). We used a radius of 25 km, thus a CA_{25} of 1 represent $25^2 \pi = 1963.5 \text{ km}^2$. We used these range size metrics, together with the number of observations for a race to classify races in terms of rarity (cf. Rabinowitz 1981; Yu and Dobson 2000). All analyses were done with the *R* environment for statistical computing (R Core Team 2014).

Diversity patterns

We computed measures of diversity for the entire country, the Mexican states (see Appendix S1 for a map of the Mexican states) and for grid cells with a $50 \times 50 \text{ km}$ (with 781 cells in Mexico) and with a $100 \times 100 \text{ km}$ cell size (194 cells in Mexico). The grid cell sizes were selected because smaller cell sizes produce many gaps that are due to absence of sampling, and not due to actual maize distribution, while larger cells generalized the patterns too much.

Diversity measures included richness (number of races), rarity-weighted richness, summed suitability, the Shannon diversity index, and the Gini coefficient (a measure of evenness). Rarity-weighted richness was computed as the average of the circular area (CA_{25}) range size for the species in an area. The Shannon index is defined as

$$H = - \sum_{i=1}^r p_i \ln p_i \quad (1)$$

where r is the total number of races and p_i is the proportion of observations of race i . We follow Jost (2006) and report e^H rather than H because e^H is more

readily interpretable. For example, a value of 10 means that the sample is equally diverse as a sample with 10 classes that all have the same proportion (complete evenness).

The Gini coefficient is defined as (Greedy and Hurn 1999):

$$G = 1 + \frac{1}{N} - \frac{2}{\mu N^2} \sum (N + 1 - i)y_i \quad (2)$$

where N is the total number of races, μ is the mean of the proportion of observations of each race, i is the position of each race indexed in non-descended order; and y_i is the proportion of observation for race i . The Gini coefficient goes from 0 to 1, where 0 corresponds with perfect equality (evenness) and 1 corresponds with perfect inequality. As our measure of evenness (E) we therefore report:

$$E = 1 - G \quad (3)$$

such that a value of 1 corresponds with perfect evenness.

Summed suitability was computed with MaxEnt (Phillips et al. 2006) species distribution models (Elith and Leathwick 2009) as implemented in *R* package *dismo* (Hijmans et al. 2013). The models predict climatic suitability for each race based on its known distribution (occurrence records in the CONABIO database). We modeled distribution for races having more than ten records, those were 47 races. We used the 19 bioclimatic variables ($\sim 1 \text{ km}^2$ spatial resolution) from the Worldclim database (Hijmans et al. 2005) as predictors. Predictions of suitability were obtained for the resolution of the bioclimatic variables; raster cells were aggregated by the median to have raster maps of 50 and 100 km resolution. Then we calculated the modeled race richness map adding individual modeled distributions.

Regions

Regions with similar races were established using hierarchical cluster analysis (“ward.D” method) based on a Bray–Curtis dissimilarity matrix (Bray and Curtis 1957) computed with *R* package ‘vegan’ (Oksanen et al. 2013). Cells with only one race were excluded. We pruned the dendrogram to 3–7 clusters and chose to retain six clusters because of their geographic consistency. We used a three by three focal

majority function to remove some of the isolated cells surrounded by another cluster.

Results

Country level patterns

The distribution of number of observations per race showed the common ‘hollow curve’ pattern of taxon abundance data (Hijmans and Spooner 2001; McGill and Collins 2003): there are a few common taxa, and there are many rare taxa (Fig. 1; Table S1 in Appendix S2). For the 59 races analyzed the evenness (E) was 0.30 and the diversity (e^H) was 23.0 ($H = 3.14$). Tuxpeño was the most common race (19 % of the observations), followed by Cónico (11 %), Comiteco (7 %), Cónico Norteño (6 %), Olotillo (6 %) and Celaya (5 %). These six races comprised 54 % of the records, while the 37 least common races (61 % of all races) were represented by only 10 % of the total number of records. Twelve races had less than 10

observations: Harinoso de Ocho, Choapaneco, Mixeño, Mixteco, Motozinteco, Mountain Yellow, Negrito, Palomero de Chihuahua, Palomero de Jalisco, Serrano, Negro de Chimaltenango and Uruapeño.

The maximum distance between two observations ($D_{\max}$) of the same race was between 0 and 2621 km. The races with a $D_{\max}$ of zero were Negro de Chimaltenango and Uruapeño (one observation each), Palomero de Jalisco (three observations) and Choapaneco (five observations). That is, even if there was more than one observation for these races, they were all from the same location. Average $D_{\max}$ was 1169 km and 34 races had a $D_{\max}$ larger than 1000 km, the five races with the largest $D_{\max}$ were: Tuxpeño (2621 km), Vandeño (2554 km), Tabloncillo (2334 km), Mushito (2291 km), and Bolita (2203 km). Seven races had a $D_{\max}$ lower than 200, indicating these have a very small area of distribution (Fig. 1; Table S1 in Appendix S2).

CA_{25} ranged from 1 to 345 and the average was 51. Nine races had a CA_{25} larger than 100, and 14 lower than 10 (Fig. 1; Table S1 in Appendix S2). The races

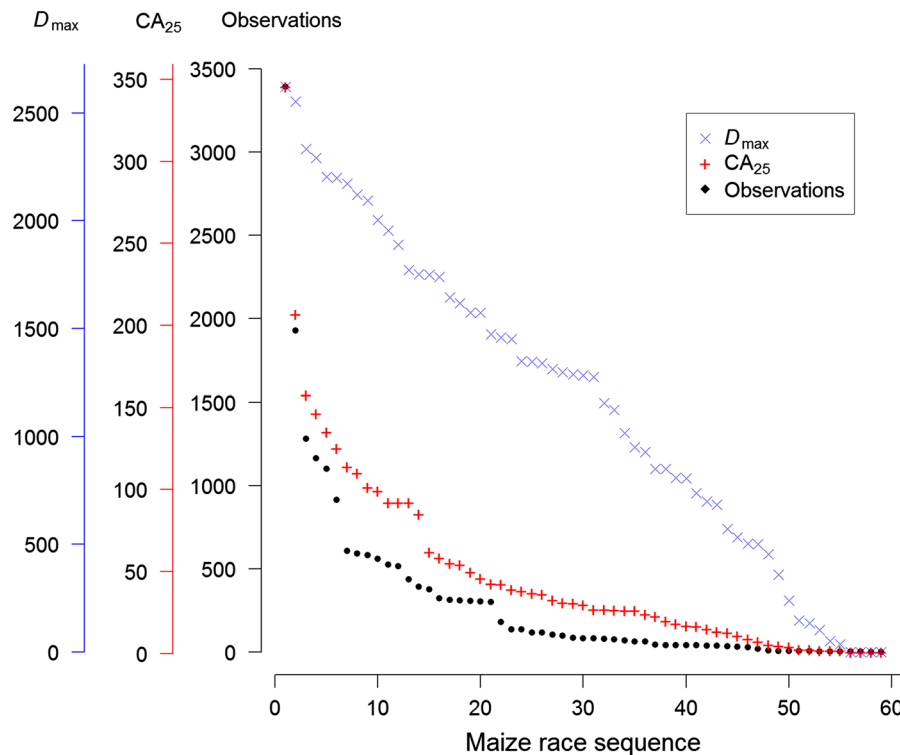


Fig. 1 Number of observations and range size ($D_{\max}$ and CA_{25}) by race. Values are ordered independently for each variable. $D_{\max}$ is the largest distance between two observations. CA_{25} is

the area covered by circles of radius 25 km around each point divided by the area of a single circle

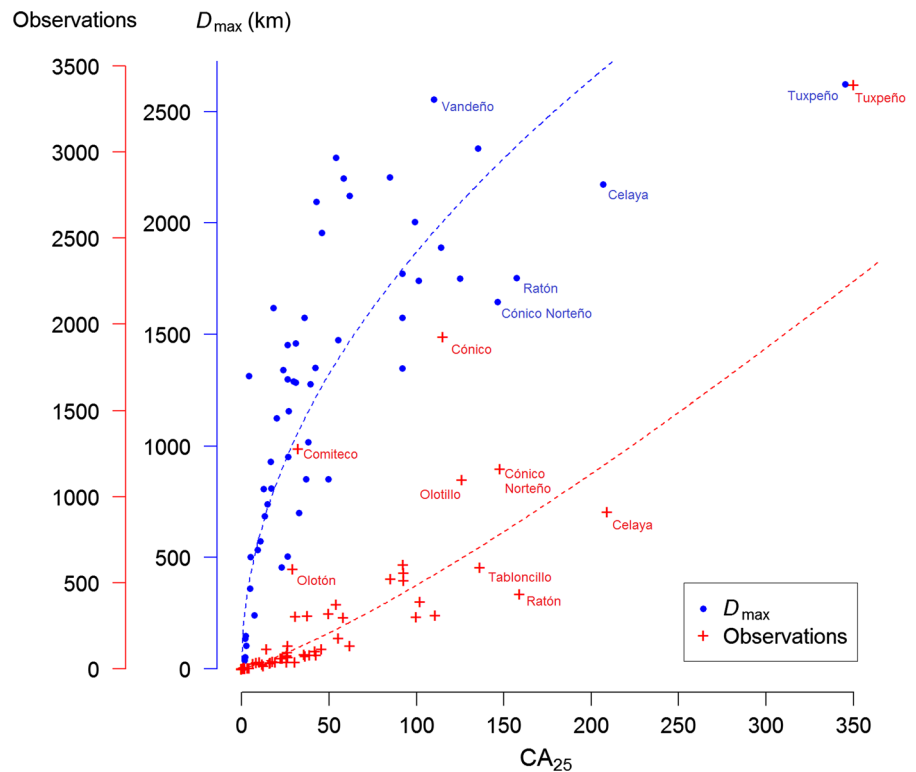
with a CA_{25} of one are those that had a D_{max} of zero. CA_{25} range size was strongly related to D_{max} and the number of observations (Fig. 2). The outliers in the relationship between CA_{25} and the number of observations are of interest as they can be used to characterize races distribution as ‘normal’, highly dispersed (relatively high CA_{25}) or strongly clustered (relatively low CA_{25}). Tuxpeño, Cónico, Comiteco and Olotón have a small range size relative to the number of observations, suggesting that they are very common in the areas in which they occur. In contrast, Ratón and Celaya had a circular area that is large relative to the number of observations, suggesting that these races are not very common in their area of distribution (Fig. 2; Table S1 in Appendix S2). Tuxeño, Celaya, Ratón and Cónico had a large CA_{25} and relative low D_{max} suggesting that their distribution is relatively compact. In contrast, Vandeño had a large D_{max} but relative low CA_{25} suggesting that it is distributed in clusters over a large area. Nine races had a relative large D_{max} , small CA_{25} and only a few observations (Table S2 in Appendix S2). Some of those races can be classified as widespread but rare, such as Azul, Cacahuacintle, Mountain Yellow,

Negrito, Palomero Toluqueño, and Zapalote Grande. Most of observations of Azul were from northern Mexico, in the state of Chihuahua but there were also two observations in the center of the country. Cacahuacintle observations were from central Mexico except for one record from Chihuahua, which dramatically increased its D_{max} . Mountain Yellow has a very low number of observations, most of them in central-west Mexico and only one in Chiapas. Nal-tel is commonly encountered in the Yucatán peninsula and elsewhere in southern Mexico, but there was one observation in Chihuahua. Negrito had very few observations, mostly in Oaxaca, but there was one in northern Veracruz. Palomero Toluqueño occurs in the highlands of the center but there was one observation in Chihuahua. Finally, Pepitilla mostly occurs in central-west Mexico but there was one observation in the southeast.

State level patterns

The original data has records for all the Mexican states (CONABIO 2013). However, for none of the observations for the state of Baja California the race name

Fig. 2 Circular area range size (CA_{25}) versus number of observations and D_{max} of Mexican maize races. Each dot refers to one race. CA_{25} is the area covered by circles with a radius of 25 km around each point divided by the area of a single circle. D_{max} is the largest distance between two observations of the same race. Regression lines are log–log for CA_{25} versus observations; and square root of CA_{25} versus D_{max}



was recorded and consequently we did not report numbers for that state. On average, there were 2.09 maize records per 100 km², but this was much higher in Tlaxcala (15.9 records per 100 km²) and Morelos (10.3). Other states with a high collection density were Chiapas (4.6), Estado de México (3.9), Puebla (2.7) and Oaxaca (2.5). Seven states had more than 20 maize races. Oaxaca and Jalisco had the highest number (28), followed by Michoacán (27), Veracruz and Durango (25), Chihuahua and Chiapas (24). The number of races in Durango and Chihuahua was high despite the low density of maize records (less than 0.5 records per 100 km²; Table S3 in Appendix S2). However, in general, the distribution of races by state follows the observed pattern in the country: there are a few very common races, and many races with very few observations. Among the states with high richness, Chiapas had the lowest evenness (0.15) and Michoacán the highest (0.49).

Gridded patterns

Observations

Grid cells with a large number of maize observations were common near the southern Mexican coast in the states of Chiapas and Oaxaca; and in Central Mexico, around Mexico City and in the state of Puebla (Figs. 3A, 4A). For the 50 km resolution raster, 34 % of the Mexican cells had no records, and 27 % had between 1 and 10 records. The highest number of records in one cell was 756.

Richness and diversity

Race richness was higher in the south than in the north (Figs. 3B, 4B), and associated with the presence of mountainous terrain (See Appendix S1, for an elevation map of Mexico). In Northern Mexico, areas of higher richness are in the west, in the Sierra Madre Occidental. There are areas of high richness in the trans-Mexican volcanic belt that crosses Mexico from the Pacific coast to the Gulf of Mexico and in the mountain ranges of Sierra Madre del Sur and Sierra de Chiapas. The cell with the highest number of races at 50 km resolution (richness = 18) was in Oaxaca.

Evenness was generally higher where richness was low and vice versa (Figs. 3C, 4C). Evenness was very low in areas of high richness in large parts of the states of

Chiapas and Oaxaca. Evenness was higher in Northern Mexico and by the Pacific coast and in the Yucatán peninsula, oftentimes because there were few observations and races in these cells. Race diversity (Figs. 3D, 4D) was strongly correlated with race richness ($R^2 = 0.86$ for 50 km and $R^2 = 0.82$ for 100 km cell size). The outliers are noteworthy: these are areas where evenness (Figs. 3C, 4C) is either very low or very high, relative to richness. Areas of high richness but low evenness occurred in Chiapas, southeast Mexico and south Oaxaca. The opposite pattern was found in central west to south Mexico (Jalisco, Michoacán, Guerrero and Oaxaca), the area where Chihuahua, Sonora and Sinaloa have a border, and a small area in the northwest (Zacatecas, Durango and Nayarit).

There was a positive association between the number of observations and the number of races at the 50 km ($R^2 = 0.55$) and 100 km cell size ($R^2 = 0.62$). The association between diversity and the number of observations was much weaker ($R^2 = 0.22$ at 50 km and $R^2 = 0.24$ at 100 km cell size).

Summed suitability

Summed suitability is an estimate of the potential (maximum expected) richness in an area. Its distribution is similar, but much smoother than observed richness (Figs. 3E, 4E). Model results show three regions of high summed suitability: a large area located in central west Mexico (Jalisco), and two small areas in southern Mexico, one in western Oaxaca and the second in central Chiapas. For the 50 km cell size grid, the maximum value of summed suitability was 24 races, very similar to the richness (18 races). For the 100 km cell size the highest value was 27 races.

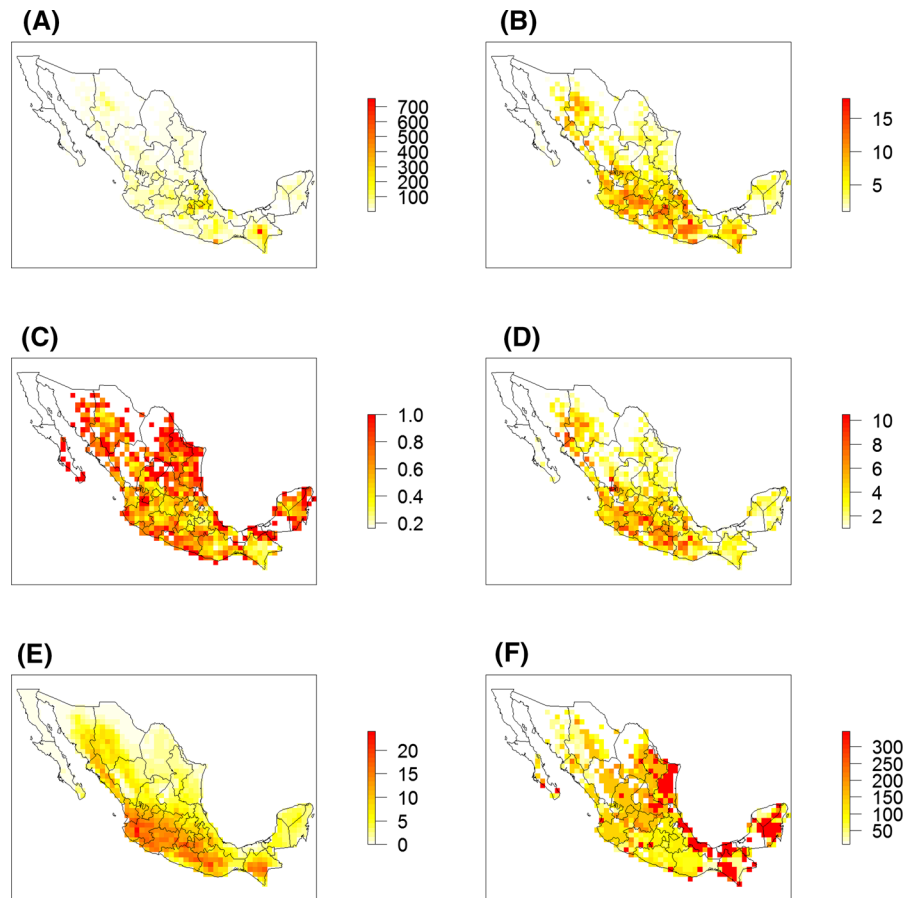
Range size

Areas in the lowlands by the coast of Gulf of Mexico have the highest larger average range size, as this is where the most widespread races such as Tuxpeño, Celaya, and Ratón occur, and some of these cells only have one race (Tuxpeño or Ratón). Low mean range size is associated with high diversity (Figs. 3F, 4F).

Regionalization

We detected six geographically consistent regions with different maize races (Fig. 5): three northern

Fig. 3 Distribution of maize races in Mexico for a 50×50 km cell size grid. **A** number of records; **B** richness; **C** evenness; **D** diversity; **E** summed suitability; **F** average range size



regions (North-Central, North-West, and North-East), one central region (Central) and two southern regions (South and South-East). The North-East region is dominated by only three races, which account for 95 % of the records, Tuxpeño (48 %), Ratón (26 %), and Tuxpeño Norteño (21 %). The North-East region includes lowlands in the northern part of the Gulf of Mexico and part of the Sierra Madre Oriental. The other northern regions have a very diverse composition having unique races with high frequency. The North-Central is a highland region located between mountain ranges and has a relatively dry and cold climate. The characteristic maize races that represent more than 80 % of the records are Cónico Norteño (30 %), Cristalino de Chihuahua (18 %), Ratón (8 %), Tuxpeño Norteño (8 %), Celaya (7 %), Tuxpeño (6 %), and Azul (6 %). The North-West region is located in the Sierra Madre Occidental, a mountain range that runs along the Pacific coast. Dominant races in this region are Tabloncillo (27 %), Tuxpeño

(18 %), Tabloncillo Perla (11 %), Reventador (5 %), Onaveño (5 %), Elotero de Sinaloa (5 %), Celaya (4 %), Vandeño (4 %), and Blando (3 %). The Central region has a complex topography composed by mountain ranges and valleys; the climate is temperate because of the elevation, and there is a north south rainfall gradient (drier in the north). Because of that, the region's race composition is diverse. The races that dominate this region are Cónico (29 %), Celaya (11 %), Cónico Norteño (11 %), Chalqueño (10 %), Elotes Cónicos (8 %), Arrocillo Amarillo (5 %), Bolita (4 %), and Tuxpeño (3 %). The South region covers a large geographic area that includes lowlands and highlands from Guerrero, Oaxaca and Chiapas with high race diversity and the lowlands of Veracruz. The dominant races are Tuxpeño (26 %), Comiteco (17 %), Olotillo (12 %), Olotón (7 %), Tepecintle (6 %), Bolita (3 %), Pepitilla (3 %), Vandeño (3 %), Ancho (3 %). The South East is located in the Yucatan Peninsula, a lowland region with hot climate. There

Fig. 4 Distribution of maize races in Mexico for a 100×100 km cell size grid. **A** number of records; **B** richness; **C** evenness; **D** diversity; **E** summed suitability; **F** average range size

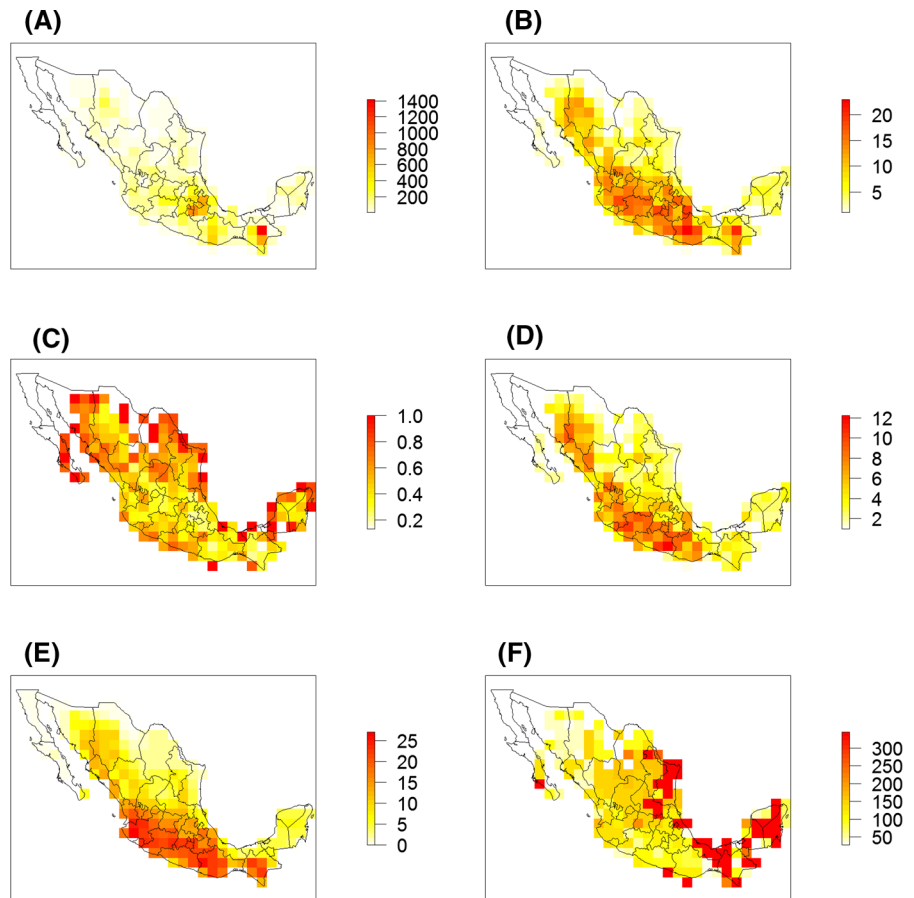
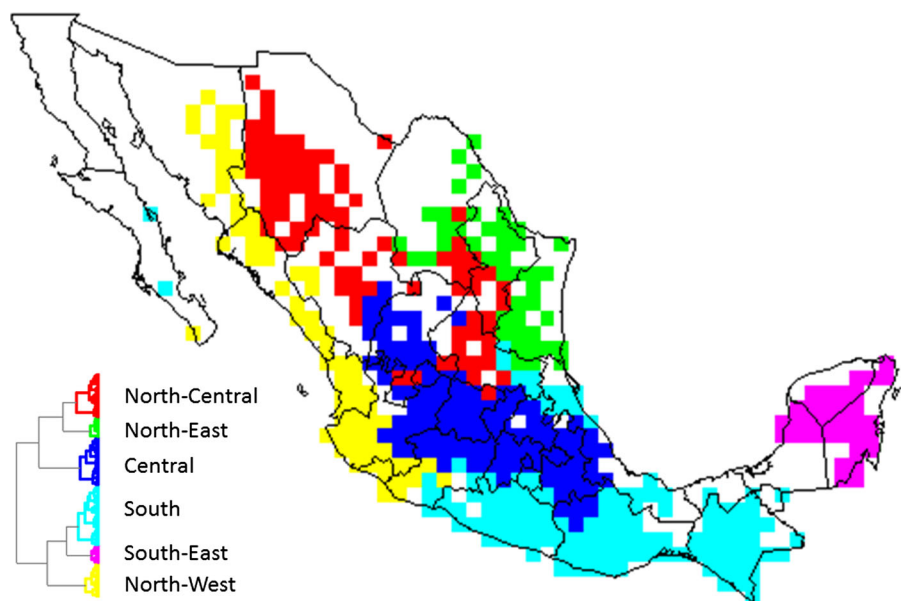


Fig. 5 Cluster dendrogram and regions based on presence-absence of maize races



the dominant races are: Tuxpeño (62 %), Olotillo (13 %), Dzit Bacal (10 %), and Nal-tel (9 %).

Discussion

Quality and completeness of the survey data

We used a uniquely large database of intra-specific crop diversity. Because of the large sample size, we believe that the data allowed us to provide a robust description of maize race diversity in Mexico. However, number of observations and range size statistics inform about possible errors probably due to race misclassifications. There are some outliers that have only a few observations, small CA_{25} and large D_{max} , that is the presence of observations that far away from the main area of distribution of a race (Table A2).

There are, however, a few discrepancies with previous studies. For example, Aragón-Cuevas et al. (2006) reported Oaxaca as the state with the largest number of races (35), whereas we report 28 races. This is in part because of differences in classification. Aragón-Cuevas et al. (2006) reported eight races in Oaxaca that are not reported for this state in the CONABIO database, including four races that do not exist in the entire database (Negro de Tierra Fría, Negro Mixteco, Olotón Imbricado, Serrano de Oaxaca), although there were two races in the CONABIO database for Oaxaca that were not reported by Aragón-Cuevas et al. (2006). Aragón-Cuevas et al. (2006) suggested further study to confirm the existence of those races.

Races with only a few observations are problematic because the concept of race is based on recurrent populations, that is, meta-populations (Anderson and Cutler 1942; Hernández and Alanís 1970). Races in this national data set that have a very low frequency should be reviewed because they may not meet the formal definition of race (Perales and Golicher 2014). Richness is also very sensitive to rare (and possibly erroneous) observations. The Shannon diversity index, in contrast, is not very sensitive to the inclusion of rare observations. Note that Michoacán, which had one race less than Oaxaca, had the highest Shannon index. The use of systematic sampling methods, instead of the common attempt to document all races present in a sampled location, could also be helpful to better estimate diversity patterns.

Although many maize races are easily recognized and well established, race is not a formal taxonomic

group. We are confident that race diversity should be a good proxy for genetic diversity. In the near future, this assumption can be tested as large scale genetic and genomic data become available for genebank collections (Prasanna 2012).

Geographic patterns of diversity

Ortega-Paczka (2003) reported that areas of high landrace richness are the central mesa (Near Toluca, Mexico), central Chiapas (Southern Mexico) and northwestern Mexico. Mera (2009) added that areas of high maize diversity are found in the Sierra Madre Occidental, the Chiapas-Oaxaca-Guatemala region, and in the Mesa Central. Perales and Golicher (2014) largely confirmed the above and proposed six distinct regions. In general terms these previous results are similar to ours, but there are some differences. Modeled richness, as used by Perales and Golicher (2014) showed high richness in almost all of the states of Jalisco, Michoacán, Guerrero, and Oaxaca. However, the empirical data suggest that there is high richness in only some parts of those states. For example, the diversity observed along the coast of Oaxaca and the northwest of Guerrero is not nearly as high as the modeled (potential) richness. Possible reasons for these discrepancies include low regional sampling densities in some areas and differences in economic and agricultural situation between regions. In the case of northwest Guerrero low sampling could explain the low observed diversity, but in coastal Oaxaca sampling has been as intense as in other regions of the state (Aragón-Cuevas et al. 2006). There, the coastal plains are used for cattle and plantation crops like coconuts and limes. The teosinte populations in Northwest Guerrero may have been the ancestor of maize (van Heerwaarden et al. 2011), and, consequently, a high maize race diversity might have been expected. More sampling in this region is needed to find out if our numbers are biased by low sample size or whether diversity is relatively low there.

Maize regions

Maize races have been grouped using morphological traits (Wellhausen et al. 1952), by isozymes (Doebley et al. 1985), by morphological characteristics and isozymes (Sanchez et al. 2000), by microsatellites (Reif et al. 2006), and by environmental variables

(Ruiz Corral et al. 2008). Racial groups (“complejos raciales”) have been proposed by grouping races according race morphological characteristics and those groups have a defined geographical distribution (Doebley et al. 1985; Sanchez et al. 2000; Reif et al. 2006). The regions that we propose here summarize individual race distribution and “maize community composition” and we think this regionalization has much to offer for understanding maize racial grouping. The regions reasonably agree with the regions proposed by Perales and Golicher (2014). Our North-West region largely coincides with their region five; our North-Central covers region six and half of region seven; our Central region corresponds to their regions four and ten; our South region covers their regions one, two, eight and half of three; and our South-East corresponds to their region nine.

It is interesting to note that most areas of high maize race richness are located in places where two or more maize regions as defined in this paper come together. For example, in the state of Jalisco, in the central-west, there is an area of high diversity, there three regions converge: North-Central, North-West and Central. In the area of high richness in Oaxaca, Central and South meet. In the north, in Chihuahua, groups North-Central, and North West come together resulting in another region of high richness. Our present paper and other recent contributions (Ruiz Corral et al. 2013; Perales and Golicher 2014) should provide a basis for more systematic and directed sampling to explore the differences between these studies, and create a robust biogeography of maize races. A practical use of this information could be to design a comprehensive plan for monitoring and in situ conservation of maize genetic resources.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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